

DIFFERENTIAL CALCULUS

THEORY

Let $y = f(x)$ be a continuous function. Then, the value of y depends upon the value of x and it changes with a change in the value of x . We use the word increment to denote a small change, i.e., increase or decrease in the values of x and y .

Let Δy be an increment in y corresponding to an increment Δx in x .

Then, $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. This limit, if it exists finitely, is called the derivative or differential coefficient of $y = f(x)$ with respect to x and is denoted by $\frac{dy}{dx}$ or $f'(x)$ or y_1 . The process of finding the derivative is known as differentiation.

Standard Derivatives

$\frac{d}{dx} x^n = n \cdot x^{n-1}$	$\frac{d}{dx} (c) = 0$	$\frac{d}{dx} x = 1$	$\frac{d}{dx} \frac{1}{x^n} = -\frac{n}{x^{n+1}}$
$\frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}$	$\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$	$\frac{d}{dx} \frac{1}{\sqrt{x}} = -\frac{1}{2x\sqrt{x}}$	$\frac{d}{dx} e^x = e^x$
$\frac{d}{dx} e^{mx} = m \cdot e^{mx}$	$\frac{d}{dx} a^x = a^x \cdot \log_e a$	$\frac{d}{dx} a^{mx} = m \cdot a^{mx} \cdot \log_e a$	$\frac{d}{dx} \log_e x = \frac{1}{x}$

Product and Quotient Rule

$$\frac{d}{dx} u \cdot v = u \cdot \frac{d}{dx} v + v \cdot \frac{d}{dx} u \qquad \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2}$$

Parametric Functions

Sometimes x and y are given as function of another variable t . Then t is called a parameter. Let $x = f(t)$ and $y = g(t)$, then:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Implicit Functions

When the variables x and y are not explicitly or clearly defined in terms of each other, the function takes an implicit form. We differentiate both sides of the equation term wise, keeping in mind that $\frac{d}{dx} 2y = 2 \cdot \frac{dy}{dx}$ & $\frac{d}{dt} m^2 = 2m \cdot \frac{dm}{dt}$ and so on.

Function of a Function – Chain Rule

If $y = f(t)$ and $t = g(x)$, then $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$, and the rule can be further extended.

Logarithmic Differentiation – Log Rule

When the given function is a power of some expression or a product of expressions, we take logarithm on both sides and differentiate the implicit functions so obtained.

If $y = f(x)^{g(x)}$, then; $\log y = g(x) \cdot \log f(x)$Then proceed.

Slope – Applied Differentiation

For $y = f(x)$, slope at any point (x_1, y_1) is given by $\frac{dy}{dx}$ at x_1, y_1

Higher Order Derivatives

Let $y = f(x)$ be a differentiable function of x whose second and higher order derivatives exists.

The first, second, third, and the n th derivatives of this function are denoted by;

$dy/dx, d^2y/dx^2, d^3y/dx^3, \dots, d^ny/dx^n$ or $y_1, y_2, y_3, \dots, y_n$ or $f'(x), f''(x), \dots$